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Project proposal

1. Describe the dataset. (ex: The number of microaneurysms found in a patient's eye and whether or not they have diabetic retinopathy; or Lending Tree loan data, including who defaulted and who paid off their loan.)

Our dataset is based on video games information sales between the 1970s and 2024. It includes. It has information like the game's title, console, genre, publisher, developer, critic score, release date, and sales across different regions of the world. We plan to use this dataset to predict whether a video game will be a high-selling title based on characteristics.

2. How many records does the dataset have?

This dataset contains 64,016 records. Only 4,123 contain all filled values for (genre, console, publisher, developer, release date, critic score, and total sales)

3. How many features does the dataset have? List or describe a few of them.

This dataset contains 14 features.

- img : URL slug for the box art at vgchartz.com
- title : Game title
- console : Console the game was released for
- genre : Genre of the game
- publisher : Publisher of the game
- developer : Developer of the game
- critic_score : Metacritic score (out of 10)
- total_sales : Global sales of copies in millions
- na_sales : North American sales of copies in millions
- jp_sales : Japanese sales of copies in millions
- pal_sales : European & African sales of copies in millions
- other_sales : Rest of world sales of copies in millions
- release_date : Date the game was released on
- last_update : Date the data was last updated

4. What can you try to predict in this dataset? (ex: We can use the number of microaneurysms measured in the patient's eye to predict whether or not they have diabetic retinopathy; or We can try using the features, including age, income, home ownership status, etc, to predict whether or not someone will default on their loan.)
We're trying to predict whether a video game becomes a high-selling game based on a variety of features (i.e. genre, console, publisher, developer, release timing and critic score), classifying "high-selling" as an arbitrary number like over 1 million copies.

5. What kind of ML problem are you doing (e,g, classification, regression, clustering)? Is this a labeled dataset, appropriate for a supervised learning classification or regression problem? (Or will your project only focus on unsupervised learning?)
We will be classifying new video games to be either high-selling or not. Therefore we will be solving a classification problem. This is not a labeled dataset since we are predicting something that is not one of the 14 features.

6. Provide a link to the dataset, if there is one. If you are getting your data from somewhere other than a link, where are you getting it from?
https://www.kaggle.com/datasets/yasrebi/video-game-sales

7. Briefly describe a plan of things you plan on doing (no need to have all details right now, but if you know your data is messy you should have a plan on how you will clean and preprocess, and also an idea of what algorithms you want to try).
Our plan is to use the algorithms KNN and Decision Trees to predict whether a new game will be a high-selling or not. We have chosen 1 million copies to be considered our high-selling. The first thing we would need to do is clean the dataset. There are irrelevant features that are not needed so we would need to remove them. We also need to convert some numerical values into categorical values. After cleaning, we will try using KNN to do predictions and Decision Trees to do predictions. We then will explain why a certain algorithm was better than the other.